DAY 14 OF 30

The data gateway

Reaching data inside a factory or office.
Without opening the firewall.

Outbound only **Clusters** **Hybrid**

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THE PROBLEM

The cloud cannot see inside your building.

1	**The database**	On a server in the plant
2	**The firewall**	Blocks everything from outside
3	**The security team**	Will not open an inbound port

And they are right. **So how does a cloud pipeline read that table?**

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THE SIMPLE VERSION

A courier who works from the inside.

The office does not let strangers walk in. Instead, an employee carries parcels out to the delivery van when asked.

The gateway is that employee: **software inside your network that fetches data on request**.

✓ **Installed**	On a machine in your network
✓ **Connects**	Outward, to the cloud
✓ **Firewall**	No inbound ports opened

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HOW IT WORKS

How a request travels

1. A pipeline or report in Fabric asks for on-premises data
2. The request waits in a **secure cloud relay**
3. The gateway **calls out**, picks it up and queries the source
4. Results travel back through the same outbound connection

Why security teams accept it

The gateway only makes outbound connections. Nothing on the internet can dial in.

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FOR ENGINEERS

Running it properly

Use standard mode

Standard mode serves many users and sources. Personal mode works only with Power BI and cannot be shared.

Build a cluster

Two or more gateway machines in a cluster avoid a single point of failure.

Keep it updated

A new version ships every month, and only the last six releases are supported.

Put it near the data

The gateway machine does real work. Place it close to the source and size it for the load.

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THE HONEST PART

Gateway or something else?

ON-PREMISES GATEWAY

Sources on servers in your own network

Strict rules against inbound access

A mix of cloud and on-premises data

CONSIDER INSTEAD

Sources in an Azure virtual network

Data already replicated to the cloud

A source that can push files out

> **The catch:** the gateway is a server someone must own. Most "the pipeline failed" tickets on hybrid projects trace back to a gateway that was offline, full or out of date.

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FROM REAL WORK

A gateway on a real project

A manufacturing group with a plant in Sweden and its analytics platform in Azure in India.

1. **On-premises gateways** connected the plant's systems
2. **Data Factory** moved the data across to Azure
3. **Legacy SQL data** landed in Synapse and Fabric
4. **Reporting was centralised** across cloud and on-premises systems

Two countries and two networks, **one reporting platform**.

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SAVE THIS

The gateway in five lines

1. **Software** inside your network
2. **Outbound only**, no open inbound ports
3. **Standard mode** and a cluster for production
4. **Monthly updates**, so plan them
5. **It needs an owner** like any server

Tomorrow: Mirroring.
A live copy of your database inside Fabric.

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